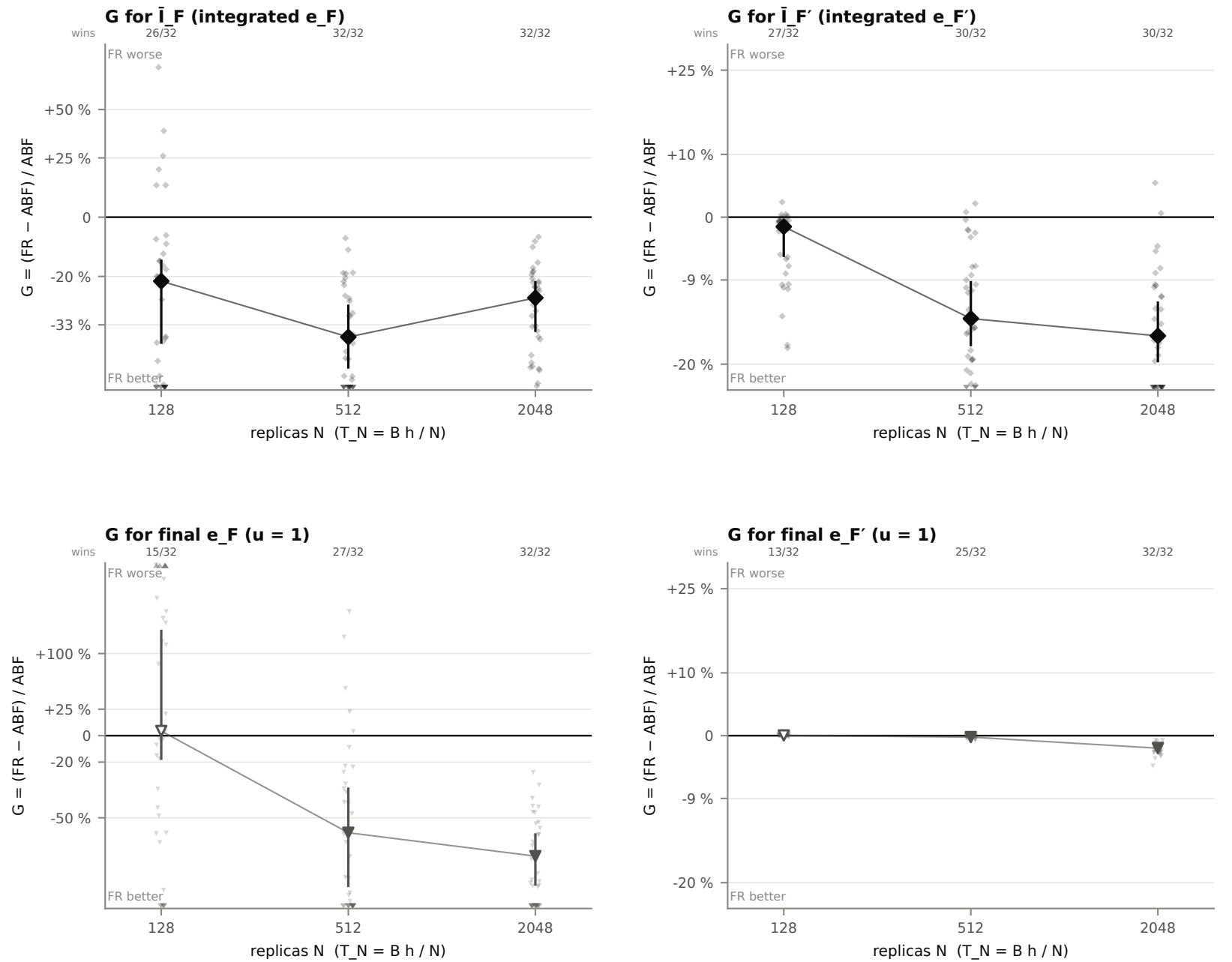


gateway alpha0.5: **FR relative improvement G(N) at equal budget**

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B\ h / N$); 32 seeds per (N , method).
Paired per seed (same N and seed: shared initial conditions and Langevin noise).



gateway alpha0.5 · S3_fr_gain_vs_N · status: all planned (N, seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design).
Marker: median of the per-seed G ; bar: seed-bootstrap 95 % CI of the median (10000 resamples); filled = CI excludes 0; faint: per-seed G (carets: off scale). Log-ratio axis (FR/ABF) labelled in % change. 'wins' row: seeds with $FR < ABF$ / paired seeds. Red 'FR inactive' / 'no death k/n': FR arms that realised no death (equal to ABF), where $G = 0$ is FR inactivity, not equivalence.
Source results/mechanism/matched_free_energy/alpha0.5/analysis/summary.json (generated 2026-10-10T14:19:19Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.